Companion X

Microscopic Derivation of the RDCC Relaxation Rate from the Dimension-Six Inter-Sector Operator

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Abstract

Relaxation-Driven Cyclic Cosmology (RDCC) describes the Universe as a globally CPT-symmetric quantum state defined on the bipartite Hilbert space $\mathcal{H}_+ \otimes \mathcal{H}_-$, with the two sectors related by CPT conjugation. The decay of inter-sector coherence is governed by the unique dimension-six interaction

$$\mathcal{L}_{\text{int}} = \frac{1}{M^2} T^{(+)}_{\mu\nu} T^{(-)\mu\nu},$$

which couples the stress tensors of the two sectors.

In this Companion we derive, from first principles, a microscopic master equation for the bipartite density matrix in an expanding FRW background. Evaluating the Born–Markov kernel with stress-tensor correlators over a Hubble-scale memory time yields a completely positive Lindblad evolution for the reduced sectoral states. The dissipative part of this evolution implies a relaxation rate

$$\Gamma(t) \propto H(t),$$

arising directly from the operator dimension, the scaling of relativistic stress tensors, and the cosmological coarse-graining set by the Hubble expansion.

This result provides the microscopic foundation for the RDCC relaxation equation $\dot{\alpha} = \Gamma\alpha$ and clarifies how decoherence, expansion, and inter-sector coupling jointly determine the evolution of the global state.

Executive Summary

This Companion provides the microscopic derivation of the RDCC relaxation rate. Starting from the unique dimension-six inter-sector interaction

$$\mathcal{L}_{\text{int}} = \frac{1}{M^2} T^{(+)}_{\mu\nu} T^{(-)\mu\nu},$$

we construct a quantum master equation for the bipartite density matrix ρ_{global} in an expanding FRW background. The derivation proceeds through a Born–Markov coarse-graining with a Hubble-scale memory time, reflecting the fact that relativistic stress-tensor correlators decay on timescales of order $H^{-1}(t)$.

Evaluating the resulting kernel yields a completely positive Lindblad evolution for the reduced sectoral states ρ_+ and ρ_-. The dissipative part of this evolution governs the decay of the off-diagonal coherence components ρ_{+-}, and the corresponding relaxation rate takes the universal form

$$\Gamma(t) \propto H(t),$$

up to dimensionless coefficients and powers of $M_{\rm Pl}/M$. This proportionality follows directly from the operator dimension, the scaling of relativistic stress tensors, and the cosmological coarse-graining implied by the expansion.

Projecting the Lindblad equation onto the coherence block yields the effective RDCC relaxation equation

$$\dot{\alpha} = \Gamma\alpha,$$

which underlies the multiplicative evolution of the coherence parameter $\alpha(a)$ and its freeze-out at late times. The microscopic derivation presented here therefore provides the structural foundation for the relaxation dynamics used throughout the RDCC framework and links decoherence, expansion, and inter-sector coupling under a single universal mechanism.

1 Introduction

Relaxation-Driven Cyclic Cosmology (RDCC) describes the Universe as a globally CPT-symmetric quantum state defined on the bipartite Hilbert space $\mathcal{H}_+ \otimes \mathcal{H}_-$. The two sectors correspond to CPT-conjugate cosmological branches with opposite thermodynamic arrows of time. Classical spacetime and sectoral thermodynamics emerge only after projection onto one branch, while the global state remains maximally symmetric at the CPT-invariant bounce.

A central dynamical ingredient of RDCC is the decay of inter-sector coherence. This process is governed by the unique dimension-six interaction

$$\mathcal{L}_{\rm int} = \frac{1}{M^2} T^{(+)}_{\mu\nu} T^{(-)\mu\nu},$$

which couples the stress-energy tensors of the two sectors. The operator is Planck-suppressed, universal, and symmetric under exchange of the sectors. Its effect is to induce a slow decay of the off-diagonal components of the global density matrix $\rho_{\rm global}$, encoded in the coherence blocks ρ_{+-} and ρ_{-+}. The phenomenological relaxation equation used throughout the RDCC framework,

$$\dot{\alpha} = \Gamma\alpha,$$

captures this decay at the coarse-grained level.

The purpose of this Companion is to derive the relaxation rate $\Gamma(t)$ microscopically. Starting from the global von Neumann equation for $\rho_{\rm global}(t)$, we construct a Born–Markov master equation in an expanding FRW background. The cosmological expansion provides a natural coarse-graining scale: stress-tensor correlators decay on timescales of order $H^{-1}(t)$, making the Hubble rate the memory scale of the effective open-system dynamics.

Evaluating the resulting kernel yields a completely positive Lindblad evolution for the reduced sectoral states ρ_+ and ρ_-. The dissipative part of this evolution governs the decay of ρ_{+-} and leads to the universal scaling

$$\Gamma(t) \propto H(t),$$

up to dimensionless coefficients and powers of $M_{\rm Pl}/M$. This proportionality follows directly from the operator dimension, the scaling of relativistic stress tensors, and the Hubble-scale memory time of the FRW background.

The microscopic derivation presented in this Companion provides the structural foundation for the RDCC relaxation dynamics, clarifies the origin of the multiplicative evolution of $\alpha(a)$, and establishes the link between decoherence, expansion, and inter-sector coupling that underlies the global architecture of RDCC.

2 Bipartite Hilbert Space and the Dimension-Six Interaction

The global RDCC state is defined on the bipartite Hilbert space

$$\mathcal{H}_+ \otimes \mathcal{H}_-,$$

where the two components correspond to CPT-conjugate cosmological branches with opposite thermodynamic arrows of time. The global density matrix $\rho_{\text{global}}(t)$ contains both sectoral states and their coherence components,

$$\rho_{\text{global}} = \begin{pmatrix} \rho_+ & \rho_{+-} \\ \rho_{-+} & \rho_- \end{pmatrix},$$

with the off-diagonal blocks ρ_{+-} and ρ_{-+} encoding inter-sector coherence. The decay of these components is the dynamical origin of the RDCC relaxation parameter $\alpha(a)$.

The only coupling between the two sectors is the unique dimension-six operator

$$\mathcal{L}_{\text{int}} = \frac{1}{M^2} T^{(+)}_{\mu\nu} T^{(-)\mu\nu},$$

which is local, covariant, CPT-symmetric, and suppressed by M^{-2}. This interaction acts on the full Hilbert space $\mathcal{H}_+ \otimes \mathcal{H}_-$ and induces weak, universal correlations between the sectors. In an FRW background with metric $ds^2 = -dt^2 + a(t)^2 d\vec{x}^2$, the corresponding interaction Hamiltonian is

$$H_{\text{int}}(t) = \frac{1}{M^2} \int d^3x \, a(t)^3 \, T^{(+)}_{\mu\nu}(t, \vec{x}) \, T^{(-)}_{\mu\nu}(t, \vec{x}), \tag{1}$$

where the factor $a(t)^3$ arises from the comoving volume element.

The full Hamiltonian governing the global evolution is

$$H(t) = H_+(t) \otimes I + I \otimes H_-(t) + H_{\text{int}}(t), \tag{2}$$

and the global density matrix satisfies the von Neumann equation

$$\dot{\rho_{\text{global}}}(t) = -i[H(t), \rho_{\text{global}}(t)]. \tag{3}$$

Because the interaction is Planck-suppressed, the weak-coupling (Born) approximation is valid throughout cosmological evolution. In the interaction picture, the density matrix obeys

$$\dot{\rho}_I(t) = -i[H_{\text{int}}(t), \rho_I(t)], \tag{4}$$

and the Born approximation allows the factorization

$$\rho_I(t - \tau) \approx \rho_+(t) \otimes \rho_-(t) \quad \text{for} \quad \tau \lesssim H^{-1}(t),$$

reflecting the fact that the interaction does not significantly disturb the short-time structure of the sectoral states.

The cosmological expansion provides the natural coarse-graining scale: stress- tensor correlators decay on timescales of order $H^{-1}(t)$, making the Hubble rate the memory scale of the effective open-system dynamics. This sets the stage for the Born–Markov master equation developed in the next section.

3 Born–Markov Master Equation in an Expanding Background

The interaction Hamiltonian $H_{\text{int}}(t)$ contains products of stress tensors from the two sectors. The corresponding second-order expansion of the interaction-picture evolution generates a universal Born–Markov kernel. This kernel governs the effective open-system dynamics of the reduced sectoral states and provides the microscopic origin of the RDCC relaxation rate.

3.1 Stress-Tensor Correlators

The Born–Markov kernel depends on correlators of the form

$$C(t,\tau) = \int d^3x\, d^3y\, a(t)^3 a(t-\tau)^3 \,\langle T^{(+)}_{\mu\nu}(t,\vec{x}) T^{(+)}_{\mu\nu}(t-\tau,\vec{y})\rangle \,\langle T^{(-)}_{\mu\nu}(t,\vec{x}) T^{(-)}_{\mu\nu}(t-\tau,\vec{y})\rangle. \tag{5}$$

Because the two sectors are CPT-conjugate and weakly coupled, the correlators factorize to leading order. For relativistic species, the dominant contribution comes from modes with physical frequencies of order $H(t)$. Faster modes dephase rapidly, while slower modes contribute negligibly to the commutator structure. The correlator therefore decays on a timescale

$$\tau \sim H^{-1}(t),$$

which sets the memory time of the effective dynamics.

3.2 Hubble-Scale Memory Time

The decay of the correlator on the Hubble timescale justifies the Markov approximation with coarse-graining interval $\Delta t \sim H^{-1}(t)$. The cosmological expansion breaks time-translation invariance on this scale, making $H^{-1}(t)$ the natural memory window for the open-system evolution. The Born–Markov kernel then reduces to

$$\dot{\rho_+}(t) = -\int_0^{H^{-1}(t)} d\tau\, \mathrm{Tr}_- \left[H_{\mathrm{int}}(t), \left[H_{\mathrm{int}}(t-\tau), \rho_+(t)\otimes\rho_-(t)\right]\right], \tag{6}$$

with an analogous equation for $\rho_-(t)$.

3.3 Lindblad Structure

The double-commutator structure ensures complete positivity once the correlators are inserted. The resulting master equation takes the Lindblad form

$$\dot{\rho_+}(t) = \sum_A \left(L_A(t)\, \rho_+(t)\, L_A^\dagger(t) - \frac{1}{2}\{L_A^\dagger(t) L_A(t),\, \rho_+(t)\}\right), \tag{7}$$

where the Lindblad operators are spatial integrals of stress-tensor components,

$$L_A(t) \sim \frac{1}{M^2}\int d^3x\, a(t)^3\, T^{(+)}_{\mu\nu}(t,\vec{x}). \tag{8}$$

The prefactor of the dissipative term is determined by the integral of the correlator over the Hubble memory window.

3.4 Scaling of the Relaxation Rate

The stress tensor of a relativistic fluid scales as

$$T_{\mu\nu} \sim M_{\mathrm{Pl}}^2 H^2,$$

so the correlator scales as

$$C(t,\tau) \sim M_{\mathrm{Pl}}^4 H^4 e^{-H\tau}.$$

Integrating over the memory window yields

$$\int_0^{H^{-1}} d\tau\, C(t,\tau) \sim M_{\mathrm{Pl}}^4 H^3.$$

Inserting this into the Lindblad prefactor gives the relaxation rate

$$\Gamma(t) \sim \frac{M_{\rm Pl}^4}{M^4} H(t), \tag{9}$$

up to dimensionless coefficients and powers of $M_{\rm Pl}/M$. The proportionality $\Gamma(t) \propto H(t)$ is therefore a structural consequence of the operator dimension, the scaling of relativistic stress tensors, and the Hubble-scale memory time of the FRW background.

This establishes the microscopic origin of the RDCC relaxation rate and prepares the ground for the projection onto the coherence components of the global density matrix in the next section.

4 Effective Lindblad and Boltzmann Equation for Coherence Decay

The Lindblad equation derived in the previous section governs the full bipartite density matrix $\rho_{\rm global}(t)$. To connect the microscopic dynamics to the RDCC relaxation equation, we now project the Lindblad evolution onto the off-diagonal coherence components ρ_{+-} and ρ_{-+}, which encode the degree of inter-sector correlation.

4.1 Sectoral Basis and Coherence Components

Let $\{|+\rangle\}$ and $\{|-\rangle\}$ denote orthonormal bases of the two sectors. In this basis, the global density matrix takes the block form

$$\rho_{\rm global} = \begin{pmatrix} \rho_+ & \rho_{+-} \\ \rho_{-+} & \rho_- \end{pmatrix},$$

where the off-diagonal blocks ρ_{+-} and ρ_{-+} quantify the coherence between the sectors. The decay of these components is the microscopic origin of the RDCC relaxation parameter $\alpha(a)$.

4.2 Projection of the Lindblad Equation

The Lindblad equation contains terms of the form

$$L_A\, \rho_{\rm global}\, L_A^\dagger, \qquad L_A^\dagger L_A\, \rho_{\rm global}, \qquad \rho_{\rm global}\, L_A^\dagger L_A,$$

with $L_A(t)$ constructed from stress-tensor components of both sectors. Projecting onto the $(+-)$ block yields an evolution equation of the form

$$\dot{\rho_{+-}}(t) = -\Gamma(t)\, \rho_{+-}(t) + i\, \Omega(t)\, \rho_{+-}(t), \tag{10}$$

where $\Gamma(t)$ is the dissipative relaxation rate and $\Omega(t)$ is a phase-shift term arising from the anti-Hermitian part of the kernel. The phase term does not affect the magnitude of ρ_{+-} and plays no role in the RDCC relaxation parameter. The physically relevant quantity is the real, dissipative rate $\Gamma(t)$.

4.3 Explicit Form of the Relaxation Rate

From the Lindblad prefactor and the correlator scaling derived in Section 3, the relaxation rate takes the form

$$\Gamma(t) \sim \frac{1}{M^4} \int_0^{H^{-1}(t)} d\tau\, C(t,\tau) \sim \frac{M_{\rm Pl}^4}{M^4} H(t), \tag{11}$$

up to dimensionless coefficients and powers of $M_{\rm Pl}/M$. The proportionality $\Gamma(t) \propto H(t)$ is therefore a direct consequence of the operator dimension, the scaling of relativistic stress tensors, and the Hubble-scale memory time.

4.4 Boltzmann-Type Coherence Equation

Defining the coherence amplitude

$$C(t) = \mathrm{Tr}(\rho_{+-}(t)),$$

the projected Lindblad equation reduces to the first-order decay equation

$$\dot{C}(t) = -\Gamma(t)\, C(t), \tag{12}$$

with solution

$$C(t) = C(t_0) \exp\left[-\int_{t_0}^{t} \Gamma(t')\, dt'\right]. \tag{13}$$

This is the microscopic origin of the RDCC relaxation equation

$$\dot{\alpha} = \Gamma\alpha,$$

which governs the multiplicative evolution of the coherence parameter $\alpha(a)$.

4.5 Interpretation

The decay of ρ_{+-} reflects the loss of inter-sector coherence induced by the dimension-six interaction. Because $\Gamma(t) \propto H(t)$, the relaxation dynamics are tied directly to the cosmological expansion rate. As the Universe expands and $H(t)$ decreases, the relaxation rate weakens and the coherence parameter $\alpha(a)$ freezes out. This behaviour underlies the late-time structure of RDCC and links decoherence, expansion, and inter-sector coupling under a single universal mechanism.

5 Microscopic Evaluation of the Relaxation Kernel

The relaxation rate $\Gamma(t)$ arises from the dissipative part of the Born–Markov kernel. In this section we evaluate the kernel explicitly and show how the proportionality $\Gamma(t) \propto H(t)$ follows from the scaling of stress-tensor correlators in an expanding FRW background.

5.1 Structure of the Kernel

The Born–Markov kernel entering the Lindblad equation has the schematic form

$$K(t) = \int_0^{H^{-1}(t)} d\tau \int d^3x\, d^3y\, a(t)^3 a(t-\tau)^3 \,\langle T^{(+)}_{\mu\nu}(t,\vec{x}) T^{(+)}_{\mu\nu}(t-\tau,\vec{y})\rangle \,\langle T^{(-)}_{\mu\nu}(t,\vec{x}) T^{(-)}_{\mu\nu}(t-\tau,\vec{y})\rangle. \tag{14}$$

The upper limit of integration is set by the Hubble memory time, reflecting the fact that stress-tensor correlators decay on timescales of order $H^{-1}(t)$. Modes with physical frequencies much larger than $H(t)$ dephase rapidly, while modes with smaller frequencies contribute negligibly to the commutator structure.

5.2 Scaling of Stress-Tensor Correlators

For a relativistic fluid in an FRW background, the stress tensor scales as

$$T_{\mu\nu} \sim M_{\mathrm{Pl}}^2 H^2.$$

The equal-time correlator therefore scales as

$$\langle T_{\mu\nu}(t) T_{\rho\sigma}(t)\rangle \sim M_{\mathrm{Pl}}^4 H^4.$$

The time-separated correlator inherits an exponential decay on the Hubble timescale,

$$\langle T(t)T(t-\tau)\rangle \sim M_{\rm Pl}^4 H^4 e^{-H\tau}, \qquad \tau \lesssim H^{-1}(t).$$

This behaviour reflects the fact that the dominant contribution comes from modes with physical frequencies of order $H(t)$.

5.3 Evaluation of the Kernel

Inserting the correlator scaling into the kernel yields

$$K(t) \sim \int_0^{H^{-1}(t)} d\tau\, a(t)^3 a(t-\tau)^3\, M_{\rm Pl}^8 H^8(t)\, e^{-2H(t)\tau}. \tag{15}$$

Over a Hubble time, the scale factor changes only by order unity, so $a(t)^3 a(t-\tau)^3 \sim a(t)^6$ up to corrections of order one. The kernel therefore scales as

$$K(t) \sim a(t)^6 M_{\rm Pl}^8 H^7(t).$$

The factor $a(t)^6$ cancels against the normalization of the Lindblad operators, which contain integrals over comoving volume. After normalization, the physical kernel entering the dissipative term becomes

$$K_{\rm phys}(t) \sim M_{\rm Pl}^8 H^7(t).$$

5.4 Relaxation Rate

The relaxation rate is obtained by inserting the kernel into the Lindblad prefactor,

$$\Gamma(t) \sim \frac{1}{M^4}\, K_{\rm phys}(t) \sim \frac{M_{\rm Pl}^8}{M^4}\, H(t), \tag{16}$$

up to dimensionless coefficients and powers of $M_{\rm Pl}/M$. The proportionality

$$\Gamma(t) \propto H(t)$$

is therefore a structural consequence of:

- the dimension-six nature of the inter-sector operator,
- the scaling $T_{\mu\nu} \sim M_{\rm Pl}^2 H^2$,
- the Hubble-scale memory time of the FRW background,
- and the exponential decay of stress-tensor correlators.

This microscopic evaluation completes the derivation of the RDCC relaxation rate and provides the quantitative link between the inter-sector interaction and the effective coherence dynamics encoded in $\dot{\alpha} = \Gamma\alpha$.

6 Discussion and Embedding into the RDCC Framework

The microscopic derivation of the relaxation rate establishes the structural origin of the RDCC relation

$$\Gamma(t) \propto H(t),$$

and clarifies how decoherence, expansion, and inter-sector coupling jointly govern the evolution of the global state. In this section we summarize the implications of this result and its role within the broader RDCC framework.

6.1 Consistency with the RDCC Relaxation Dynamics

The proportionality $\Gamma(t) \propto H(t)$ provides the microscopic foundation for the RDCC relaxation equation

$$\dot{\alpha} = \Gamma\alpha,$$

which governs the multiplicative evolution of the coherence parameter $\alpha(a)$. Because the relaxation rate decreases proportionally to the Hubble rate, the coherence parameter naturally freezes out as the Universe expands. This behaviour underlies the late-time structure of RDCC and explains why $\alpha(a)$ becomes effectively constant in the infrared.

6.2 Connection to Earlier Companions

The result derived here is consistent with the dynamical and phenomenological structure developed in earlier Companions:

- The freeze-out of $\alpha(a)$ (Companion VI) follows from the decay of $\Gamma(t)$ as $H(t)$ decreases.
- The log-periodic tensor modulation (Companions II–V) arises from interference between the two sectors, whose coherence is governed by the same relaxation rate.
- The stability of the RDCC prediction vector (Companion VII) reflects the universality of the relaxation dynamics and their insensitivity to microscopic details.

The microscopic derivation therefore completes the dynamical picture developed in the earlier Companions and provides a unified origin for the relaxation process.

6.3 Interpretation of the Dimension-Six Operator

The operator $\mathcal{L}_{\text{int}} = \frac{1}{M^2} T^{(+)}_{\mu\nu} T^{(-)\mu\nu}$ plays a dual structural role:

- It governs the decay of inter-sector coherence through the Lindblad kernel.
- It induces the interference pattern responsible for tensor-sector modulation.

Both effects arise from the same stress-tensor correlator structure. The dimension-six operator therefore unifies the relaxation dynamics and the tensor phenomenology under a single microscopic mechanism.

6.4 Emergence of Macroscopic Structure

The relation $\Gamma(t) \propto H(t)$ implies that decoherence is driven by the cosmological expansion itself. As the Universe expands, the Hubble rate sets the timescale on which inter-sector coherence decays. This provides a natural explanation for the emergence of classical spacetime, causal structure, and sectoral thermodynamics: the expansion continuously suppresses coherence between the sectors, making the global state effectively classical on Hubble timescales.

6.5 Limitations and Future Directions

The derivation presented here relies on:

- the Born approximation (weak coupling),
- the Markov approximation (Hubble-scale memory time),

- and the scaling of stress-tensor correlators for relativistic species.

Extensions of this analysis include:

- non-Markovian corrections near the bounce,
- mode-dependent relaxation rates,
- and a full information-theoretic treatment of inter-sector entanglement.

These refinements may reveal additional structure in the relaxation kernel, particularly in regimes where the Hubble rate varies rapidly.

The microscopic derivation of $\Gamma(t)$ presented in this Companion therefore provides the structural link between the dimension-six operator and the phenomenological relaxation dynamics used throughout the RDCC framework, and it sets the stage for the non-Markovian refinement developed in the next section.

7 Non-Markovian Memory Window at the Bounce

The Born–Markov approximation relies on the existence of a finite memory timescale set by the Hubble rate. Throughout the expanding and contracting phases, the Hubble time $H^{-1}(t)$ provides a natural coarse-graining window because stress-tensor correlators decay on this timescale. At the CPT-symmetric bounce, however, the Hubble rate vanishes,

$$H(t_b) = 0,$$

and the Markov approximation formally breaks down. In this section we quantify this breakdown and derive the leading non-Markovian correction to the relaxation kernel.

7.1 Breakdown of the Markov Approximation at $H = 0$

The Markov approximation assumes that the correlator $C(t, \tau)$ decays on a timescale $\tau \sim H^{-1}(t)$. At the bounce, the physical Hubble rate vanishes and the memory time diverges. The correlator therefore retains finite temporal support around the bounce, and the replacement

$$\int_0^\infty d\tau\, C(t, \tau) \;\to\; C(t, 0)\, H^{-1}(t)$$

is no longer valid. Instead, the kernel acquires a compact, symmetric memory window determined by the curvature of the scale factor near the bounce.

Expanding the scale factor around the bounce,

$$a(t) = a_b + \frac{1}{2}\,\ddot{a}_b(t - t_b)^2 + \mathcal{O}((t - t_b)^3),$$

with $\ddot{a}_b > 0$ for a non-singular bounce, the physical frequencies of stress-tensor modes vary quadratically in $(t - t_b)$. The correlator decays on a timescale

$$\Delta t_{\rm mem} \sim \ddot{a}_b^{-1/2},$$

which replaces the divergent Hubble time.

7.2 Minimal Non-Markovian Kernel

To capture the leading correction, we introduce a Gaussian memory kernel

$$K_{\rm NM}(t,\tau) = \exp\left[-\frac{\tau^2}{2\,\Delta t_{\rm mem}^2}\right],$$

which reduces to the Markovian delta kernel in the limit $\Delta t_{\rm mem} \to 0$. The Born expansion of the interaction-picture evolution then yields

$$\dot{\rho}_I(t) = -\frac{1}{M^4}\int_0^\infty d\tau\, K_{\rm NM}(t,\tau)\,[T(t),[T(t-\tau),\rho_I(t)]]. \tag{17}$$

This equation is completely positive and reduces to the Lindblad form away from the bounce. The non-Markovian correction is confined to a narrow interval $|t-t_b| \lesssim \Delta t_{\rm mem}$.

7.3 Correction to the Relaxation Rate

Projecting onto the coherence block ρ_{+-} yields a modified relaxation equation

$$\dot{\rho_{+-}}(t) = -\Gamma_{\rm eff}(t)\,\rho_{+-}(t),$$

with effective rate

$$\Gamma_{\rm eff}(t) = \Gamma_{\rm Markov}(t)\left[1+\varepsilon_{\rm NM}\exp\left(-\frac{(t-t_b)^2}{2\,\Delta t_{\rm mem}^2}\right)\right], \tag{18}$$

where $\varepsilon_{\rm NM}$ is a dimensionless coefficient determined by the curvature of the scale factor and the correlator structure. The correction is localized at the bounce and integrates to a finite shift in the initial coherence amplitude.

7.4 Physical Interpretation

The non-Markovian memory window has three structural implications:

- It regularizes the Markovian divergence at $H(t_b) = 0$ and yields a finite, symmetric initial relaxation kernel.
- It provides a controlled mechanism for generating the primordial coherence structure of the CPT-conjugate sector.
- It preserves global CPT symmetry: the memory window is symmetric under $t - t_b \mapsto -(t-t_b)$.

Away from the bounce, the standard RDCC scaling $\Gamma(t) \propto H(t)$ is recovered exactly. The non-Markovian correction therefore acts as a localized, structural refinement of the microscopic relaxation dynamics without modifying their large-scale behaviour.

8 Outlook

The microscopic derivation of the relaxation rate establishes the structural origin of the RDCC relation $\Gamma(t) \propto H(t)$. This result unifies the relaxation dynamics, the emergence of classical spacetime, and the tensor-sector phenomenology under a single mechanism governed by the dimension-six inter-sector operator. The analysis presented here completes the dynamical foundation of RDCC and clarifies how decoherence, expansion, and inter-sector coupling jointly determine the evolution of the global state.

Several directions follow naturally from this work:

- **Information-theoretic formulation.** A full treatment of inter-sector entanglement entropy and its evolution under $\mathcal{L}_{\text{int}} = \frac{1}{M^2} T^{(+)}_{\mu\nu} T^{(-)\mu\nu}$ may reveal deeper connections between decoherence, causal structure, and emergent geometry.

- **Tensor-sector phenomenology.** The same correlators that govern relaxation also shape the log-periodic modulation of primordial tensor modes. Future gravitational-wave observations may therefore probe the microscopic coherence structure of the RDCC state.

- **Non-Markovian refinements.** The breakdown of the Markov approximation near the bounce suggests additional structure in the relaxation kernel. A systematic treatment of non-Markovian effects may refine the early-time behaviour of $\Gamma(t)$ and its role in setting initial coherence conditions.

- **Numerical implementation.** Incorporating the relaxation kernel into a Boltzmann code would allow for global parameter scans and quantitative comparison with cosmological data, enabling a full observational assessment of RDCC.

- **Emergent geometry.** The relation $\Gamma(t) \propto H(t)$ suggests that aspects of large-scale geometry may be understood in terms of correlation decay rather than fundamental spacetime primitives, providing a potential bridge between RDCC and information-theoretic approaches to gravity.

The results of this Companion provide the microscopic foundation for the relaxation dynamics used throughout the RDCC framework and set the stage for further developments in the information-theoretic and phenomenological structure of cyclic cosmology.

Appendix A: Projective Compactification of the CPT Fixed Point

The global RDCC state lives on the bipartite Hilbert space $\mathcal{H}_+ \otimes \mathcal{H}_-$, with the two sectors related by global CPT symmetry. At the CPT-invariant bounce, the coherence parameter satisfies $\alpha(t_b) \to 0$, the two sectors become maximally entangled, and the coarse-grained entropy of the global state reaches its minimum. This hypersurface is the structural fixed point of the CPT symmetry. In this appendix we formalize this fixed point using a projective compactification of the time coordinate.

Projective Structure

We introduce the real projective line $\mathbb{RP}^1$, the one-point compactification of $\mathbb{R}$ in which $+\infty$ and $-\infty$ are identified. Points on $\mathbb{RP}^1$ are equivalence classes $[x : y]$ with $[x : y] = [\lambda x : \lambda y]$ for $\lambda \neq 0$. The ordinary reals embed via $t \mapsto [t : 1]$, while the projective point at infinity is $[1 : 0]$.

To describe the neighbourhood of the bounce, we define a projective deviation coordinate

$$\tau = \frac{1}{a - a_b},$$

with $a_b = 1$ the minimal scale factor. As $a \to 1^+$ (the $+$ sector), one has $\tau \to +\infty$; as $a \to 1^-$ (the $-$ sector), one has $\tau \to -\infty$. Both limits map to the same projective point:

$$\lim_{\tau \to \pm\infty} [1 : 1/\tau] = [1 : 0].$$

Thus the bounce corresponds to the projective point where the two time directions are identified.

CPT Action

In these coordinates, the global CPT transformation acts as

$$\tau \mapsto -\tau,$$

enforcing

$$\rho_{+}(+\tau) = \Theta\, \rho_{-}(-\tau)\, \Theta^{-1},$$

where Θ denotes the CPT operator. At the fixed point $\tau = 0$, the two reduced density matrices coincide and the entanglement entropy is maximal, consistent with $\alpha(t_b) = 0$.

Geometric Interpretation of Relaxation

The multiplicative evolution of the coherence parameter,

$$\dot{\alpha} = \Gamma\alpha, \qquad \Gamma(t) \propto H(t),$$

describes the flow of $\alpha(a)$ away from the projective fixed point. As the Universe moves along the compactified line, the distance from $\tau = 0$ increases and the two sectors decohere at a rate set by the Hubble expansion. The projective compactification therefore provides a geometric interpretation of the relaxation dynamics: the bounce is a removable singularity in the global state, and the two thermodynamic arrows emerge as opposite directions along the projective line.

Coherence Length

The characteristic coherence length

$$\ell_{\rm coh} \sim \frac{c}{H(t)}$$

emerges naturally as the scale controlling how rapidly correlations can propagate across the projective fixed point before relaxation enforces sectoral projection. This geometric picture complements the dynamical and microscopic derivations presented in the main text and clarifies the structural unity of the RDCC framework.